

Slide 1 – Title Slide

Docker: The Foundation of Modern DevOps & MLOps

Mini Project – MLOps Course

Your Name | University | Semester

Slide 2 – Introduction: What is Docker?

- Open-source platform for building, packaging, and running applications in containers
- Containers include code, dependencies, and libraries
- Enables consistency and portability across environments

Slide 3 – Why Containers Matter

- Portable: Run anywhere
- Consistent: Eliminates “works on my machine” issues
- Fast: Start in milliseconds
- Isolated: Each app has its own environment

Slide 4 – Docker in Modern Engineering

Docker empowers teams with:

- Reliable deployments
- Environment reproducibility
- Scalability in cloud environments
- Seamless CI/CD pipelines

Slide 5 – Key Components

- Containerization Engine – Runs isolated apps
- Docker Images – Reusable, versioned templates
- Docker Containers – Fast runtime instances
- Docker Hub – Cloud registry for images
- Docker Compose – Multi-container orchestration
- Portability Across Clouds – AWS, Azure, GCP

Slide 6 – Cost and Availability

Component — Cost
Docker Engine (Linux) — Free
Docker CLI — Free
Docker Hub (basic) — Free
Docker Desktop — Free (for individuals/students)
Pro / Team / Business — Paid

100% Free for students and academic projects.

Slide 7 – Docker in the DevOps Lifecycle

- Plan: Define container architecture
- Build: Write Dockerfile, create image
- Test: Automated tests in containers
- Release: CI/CD pipelines push/pull images
- Deploy: Run on Kubernetes or cloud services
- Operate: Containers host microservices
- Monitor: Prometheus, Grafana, Datadog

Slide 8 – Advantages

- Portability
- Reproducibility
- Lightweight & fast
- Isolation
- Rapid deployment
- Massive ecosystem
- Multicloud support

Slide 9 – Limitations

- Requires learning curve
- Not full OS virtualization
- Shared kernel → less isolation
- Docker Desktop can be resource-heavy
- Needs Kubernetes for large-scale systems

Slide 10 – Field Applications

- Machine Learning / MLOps: Reproducible training, model serving
- Web Development: Backend, DBs, APIs
- Enterprise / Microservices: Isolated services
- Cloud Platforms: Core of AWS ECS, GCP, Azure
- Education & Research: Reproducible environments

Slide 11 – Conclusion

Docker revolutionized DevOps & MLOps.
It enables fast, reproducible, portable applications.
Essential for ML model deployment, CI/CD, and cloud-native systems.

“Docker didn’t just change how we deploy apps — it changed how we think about environments.”